

Plate Boundaries and the Landforms They Create

PhET Plate Tectonics · PhET

Year 8

~30 minutes

Science · Earth & Space Science

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/plate-tectonics>

Curriculum links: AC9S8U03 · AC9S8U04 · plate tectonics

Learning intentions

- I can describe what happens at convergent, divergent and transform plate boundaries.
- I can predict the landform produced at each type of boundary before observing it in the simulation.
- I can connect the movement at a boundary to the type of rock formed there.

Getting started

Open PhET Plate Tectonics and select the cross-section view. Start with a convergent boundary and read the labels on the plates before you drag anything.

Tasks

1. Set the simulation to a divergent boundary. Predict what landform will form as the plates pull apart, then drag the plates and check your prediction.

2. Switch to a convergent boundary. What happens to the two plates as they move together? Describe the landform that results.

3. Test all three boundary types and record the plate movement and resulting landform for each.

Boundary type	Plate movement	Landform produced

4. Compare what happens at a convergent boundary with what happens at a transform boundary. Use the words 'friction' and 'landform' in your explanation.

5. At the convergent boundary, one plate is pushed down beneath the other. Based on what you observed, what type of rock (igneous, sedimentary or metamorphic) would you expect to form from the melted material at this boundary, and why?

6. Explain how the different landforms you observed at each boundary type provide evidence for the theory of plate tectonics.

7. Identify a real-world example of one of these boundary types that you may have learned about in class, and describe the landform found there.

Extension (optional)

Mount Everest and deep ocean trenches are both found near plate boundaries but look completely different. Use what you learned about boundary types to explain why.
